Library Catalog Lookup

Practice Part A - Problem 1 | Python

```
Task: Binary-search the sorted catalog by ISBN. Compare ISBN strings so leading zeroes remain.
def findBook(catalog, targetIsbn):
low, high = 0, len(catalog) - 1
while low <= high:
mid = (low + high) // 2
isbn, title = catalog[mid]
if isbn == targetIsbn:
return title
if isbn < targetIsbn:
low = mid + 1
else:
high = mid - 1
return 'Not Found'

catalog = [('0001112223','Introduction to Algebra'),
('0002223334','Beginning Python'),
('0003334445','Classic Mythology'),
('0004445556','Data and Society'),
('0005556667','European History')]
print(findBook(catalog, '0003334445'))
print(findBook(catalog, '0009998887'))

Output: Classic Mythology; Not Found.
Complexity: O(log n) time and O(1) extra space.
```